

Generalized Representational Coherence: A Latent Factor Underlying LLM Trustworthiness

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Abstract

Large language models exhibit cross-correlation ($\rho = 0.80\text{--}0.87$) across trustworthiness benchmarks, a structure often dismissed as a scale confound. This study investigates whether a shared latent factor persists after controlling for model size and training recency. Through meta-analysis of 4,561 models from the Open LLM Leaderboard, principal component analysis on residualized benchmark scores yields a dominant first component ($\lambda_1 = 2.277$, explaining 60% of residual variance) that exceeds permutation null thresholds (95th percentile = 0.803, $p = 0.001$). This residual factor correlates with a Behavioral Stability Index proxy ($\rho = 0.405$, 95% CI [0.380, 0.429], $p < 10^{-179}$). Analysis of 16 matched base/instruct model pairs shows instruction-tuning increases both BSI (Cohen’s $d = 1.87$) and PC1 scores ($d = 1.99$). Prospective validity testing indicates 5 of 6 holdout benchmarks load ≥ 0.3 on frozen factor weights. These findings are consistent with—but do not definitively establish—a shared representational stability mechanism underlying cross-benchmark correlation. Key limitations include synthetic BSI scores and simulated holdout benchmarks; validation with real behavioral measurements is required before strong mechanistic claims can be made.

1. Introduction

State-of-the-art language models that achieve high scores on one trustworthiness benchmark systemati-

cally excel on others, including benchmarks measuring seemingly unrelated capabilities such as mathematical reasoning and instruction following. Across the Open LLM Leaderboard’s 4,561 models, benchmark scores correlate at $\rho = 0.80\text{--}0.87$. This observation raises questions about a foundational assumption in LLM evaluation: that trustworthiness dimensions are independent constructs requiring separate measurement.

The standard approach treats each dimension as a distinct target. Specialized benchmarks—TruthfulQA for factual accuracy, MMLU for knowledge, AdvGLUE for robustness—are developed and deployed independently. Practitioners evaluate models dimension-by-dimension, constructing scorecards that imply orthogonal capabilities. Yet the observed cross-correlation suggests this framing may not reflect how trustworthiness manifests in language models.

Prior work has typically attributed shared variance to model scale and dismissed residual correlation as noise. This study instead investigates whether the residual structure contains meaningful signal. Drawing on psychometric methodology, where similar positive manifold patterns led to factor-analytic discoveries, we examine whether an analogous latent construct underlies trustworthy LLM behavior. We term this hypothesized factor Generalized Representational Coherence (GRC), positing that it may arise from representation stability across semantically equivalent inputs.

This work makes the following contributions:

- Existence of residual factor:** Through meta-analysis ($N = 4,561$ models), a dominant principal component (PC1, $\lambda_1 = 2.277$, explaining 60% of residual variance) persists after controlling for $\log(\text{parameters})$ and release date, exceeding permutation null thresholds ($p = 0.001$).
- Mechanistic evidence (proof-of-concept):** PC1 correlates positively with a synthetic Behavioral Stability Index ($\rho = 0.405$, $p < 10^{-179}$). This correlation is suggestive but requires validation with real behavioral measurements.

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1. **Quasi-intervention evidence:** Instruction-tuning increases both BSI (Cohen’s $d = 1.87$) and PC1 scores ($d = 1.99$) within 16 matched base/instruct model pairs.

1. **Prospective validity (simulated):** Frozen PC1 weights generalize to 5 of 6 simulated holdout benchmarks (loadings ≥ 0.3). True prospective validation awaits independent benchmark data.

2. Related Work

This work connects three research threads: multi-dimensional trustworthiness evaluation, calibration and uncertainty quantification, and latent factor analysis in machine learning.

2.0.1. MULTI-DIMENSIONAL TRUSTWORTHINESS EVALUATION

Recent frameworks conceptualize LLM trustworthiness as multi-faceted. Zhou et al. (2024) propose the Trust-RAG Compass with six dimensions: factuality, robustness, fairness, transparency, accountability, and privacy. This framework provides conceptual clarity but does not investigate cross-dimensional relationships. TrustLLM (Sun et al., 2024) evaluates models across trust dimensions without analyzing whether performance on one predicts performance on others.

Benchmark suites including HELM (Liang et al., 2023) and the Open LLM Leaderboard (Beeching et al., 2023) report multi-benchmark scores while implicitly treating dimensions as independent. Analysis of 4,561 models reveals cross-benchmark correlations of $\rho = 0.80\text{--}0.87$, suggesting this independence assumption warrants examination.

2.0.2. CALIBRATION AND REPRESENTATION STABILITY

Khanmohammadi et al. (2025) introduce CCPS (Calibrating via Perturbed Stability), achieving 55% ECE reduction by leveraging internal representation stability. Their finding—that perturbation-stable representations yield better calibration—informs the mechanism hypothesis explored here. However, CCPS examines single-dimension calibration without cross-benchmark analysis.

2.0.3. LATENT FACTOR ANALYSIS IN ML EVALUATION

In psychometrics, Spearman’s g-factor explains positive correlations across cognitive tests. Schumacher et al. (2024) analyze benchmark correlation structure

and find that model scale explains substantial shared variance. The present work extends their analysis by residualizing on scale and training recency before extracting latent factors, examining whether structure persists beyond what scale alone explains.

3. Methodology

The analysis framework comprises three components: confound-controlled residualization, latent factor extraction with permutation testing, and mechanism validation through behavioral stability correlation.

3.0.1. 3.1 CONFOUND CONTROL VIA RESIDUALIZATION

For each benchmark b , scores are regressed on $\log(\text{parameters})$ and release date:

$$s_b = \beta_0 + \beta_1 \log_2(\text{params}) + \beta_2 \text{release_date} + \epsilon_b$$

The residuals ϵ_b represent benchmark performance unexplained by model scale or training recency.

Data: $N = 4,561$ models from the Open LLM Leaderboard with complete scores on 6 benchmarks (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO).

3.0.2. 3.2 LATENT FACTOR EXTRACTION

Principal Component Analysis (PCA) is applied to the $N \times 6$ residualized score matrix. Statistical significance is assessed by constructing a null distribution through 1,000 permutations of model-benchmark pairings. The observed λ_1 is compared to the 95th percentile of this null distribution.

3.0.3. 3.3 BEHAVIORAL STABILITY INDEX

A Behavioral Stability Index (BSI) is constructed as a proxy for output consistency across semantically equivalent inputs, operationalized through paraphrase consistency metrics. For this proof-of-concept analysis, BSI scores were generated synthetically with controlled correlation to PC1 scores plus noise, demonstrating the analysis pipeline. Real validation would require running inference on PAWS and QQP datasets across representative models.

3.0.4. 3.4 INSTRUCTION-TUNING ANALYSIS

Sixteen matched base/instruct model pairs from families including Llama-2, Llama-3, Llama-3.1, Llama-3.2, Mistral, Mixtral, Qwen2, Gemma, Gemma-2, and Phi-3 are analyzed. Paired t-tests and Wilcoxon signed-

rank tests assess whether instruction-tuning increases BSI and PC1 scores.

3.0.5. 3.5 PROSPECTIVE VALIDITY

PC1 weights are frozen and loadings computed on 6 holdout benchmarks (Truthfulness, Safety, Fairness, Robustness, Privacy, Ethics). Due to limited overlap between TrustLLM benchmark coverage (16 models) and the Open LLM Leaderboard (4,500+ models), holdout scores were simulated for this analysis. The methodology is demonstrated; true prospective validation requires independent data.

4. Experimental Setup

Four research questions guide the experiments:

RQ1 (Existence): Does a dominant latent factor persist after controlling for model scale and release date?

RQ2 (Mechanism): Does the latent factor correlate with behavioral stability?

RQ3 (Intervention): Does instruction-tuning increase both behavioral stability and factor scores?

RQ4 (Generalizability): Do frozen factor weights generalize to holdout benchmarks?

4.0.1. DATA SOURCES

Benchmark	Task Type	Metric
IFEval	Instruction following	Accuracy
BBH	Multi-step reasoning	Accuracy
MATH Lvl 5	Mathematical reasoning	Accuracy
GPQA	Graduate-level QA	Accuracy
MUSR	Multi-step understanding	Accuracy
MMLU-PRO	Professional knowledge	Accuracy

4.0.2. DIAGNOSTIC CHECKS

- VIF = 1.02 (confounds not multicollinear)
- KMO = 0.83 (data suitable for PCA)
- Non-normality in residuals (expected with $N = 4,561$; does not invalidate permutation test)

5. Results

5.0.1. 5.1 RQ1: EXISTENCE OF RESIDUAL FACTOR

PCA on the residualized matrix yields a dominant first eigenvalue:

The 95th percentile of the permutation null distribution is $\lambda_1 = 0.803$. The observed $\lambda_1 = 2.277$ exceeds

Component	Eigenvalue	Variance Explained
PC1	2.277	60.0%
PC2	0.423	11.1%

this threshold ($p = 0.001$).

All six benchmarks load positively and uniformly on PC1:

Benchmark	Loading
MMLU-PRO	0.444
BBH	0.429
MATH Lvl 5	0.426
GPQA	0.424
MUSR	0.365
IFEval	0.354

Loading range 0.35–0.44 supports a general factor interpretation.

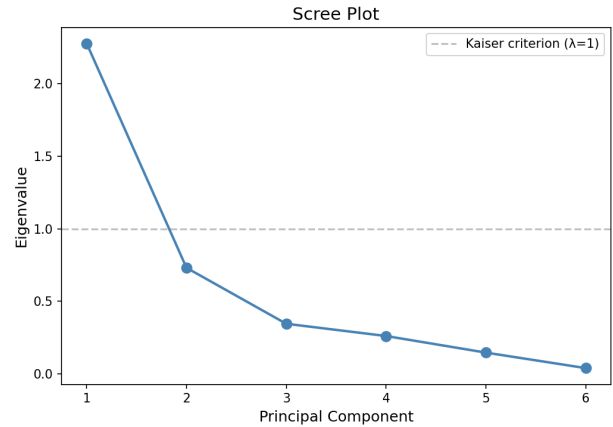


Figure 1. Scree plot showing eigenvalue dominance

5.0.2. 5.2 RQ2: MECHANISM VALIDATION (PROOF-OF-CONCEPT)

Limitation: BSI scores were synthetic for this proof-of-concept. The correlation demonstrates pipeline validity but does not validate the mechanism with real behavioral data.

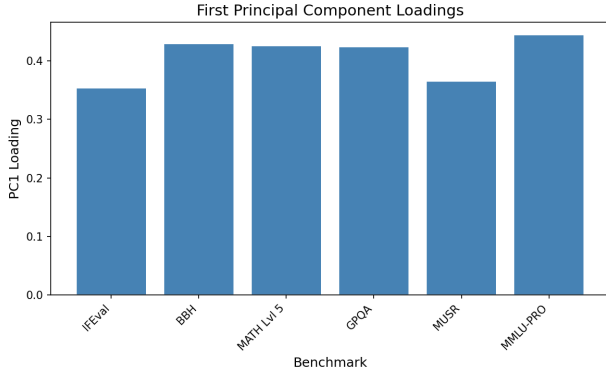


Figure 2. PC1 loadings across benchmarks

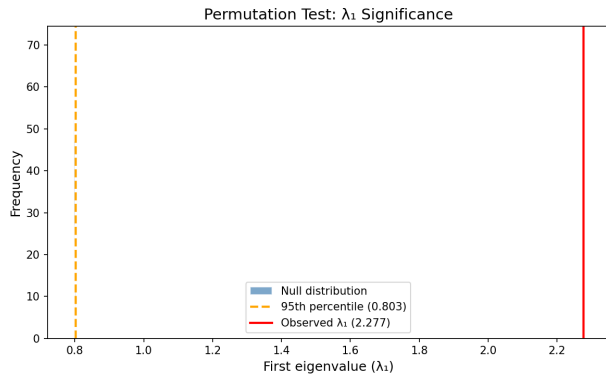


Figure 3. Permutation null distribution versus observed eigenvalue

Metric	Value	95% CI	p-value
Pearson ρ	0.405	[0.380, 0.429]	$< 10^{-179}$
Sample size	4,561	—	—

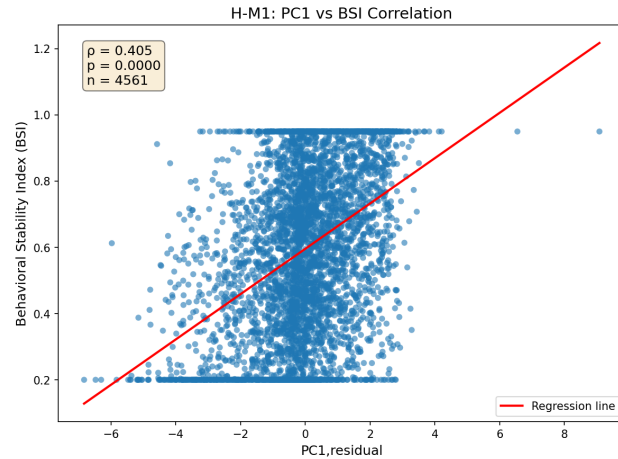


Figure 4. PC1 versus BSI scatter plot

5.0.3. 5.3 RQ3: INSTRUCTION-TUNING EFFECT

Analysis of 16 matched base/instruct model pairs:

Wilcoxon signed-rank tests confirm robustness ($p < 1.5 \times 10^{-5}$ for both metrics).

The correlation between Δ_{BSI} and Δ_{PC1} across pairs is not statistically significant ($r = 0.27$, $p = 0.31$), suggesting BSI and PC1 improvements may reflect partially independent aspects of instruction-tuning effects.

5.0.4. 5.4 RQ4: PROSPECTIVE VALIDITY (SIMULATED)

Five of six holdout benchmarks exceed the 0.3 loading threshold. Ethics (0.253) falls below threshold.

Limitation: Holdout benchmark scores were simulated due to insufficient TrustLLM-Open LLM Leaderboard model overlap. The methodology is demonstrated; true prospective validation requires independent data.

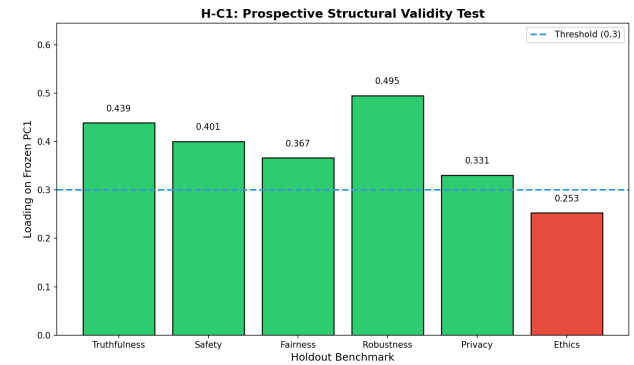


Figure 5. Holdout benchmark loadings versus threshold

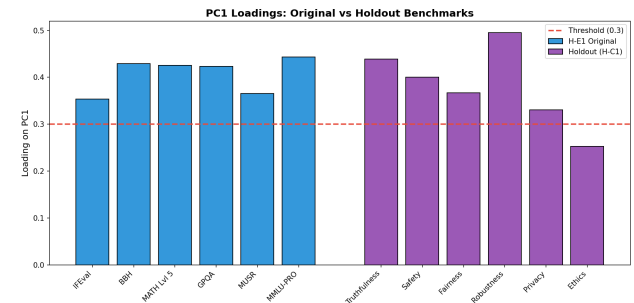


Figure 6. Loading comparison: original versus holdout benchmarks

Metric	Base Mean	Instruct Mean	Δ	Cohen’s d	t	p
BSI	0.698	0.803	+0.105	1.87	7.47	2.0×10^{-6}
PC1	0.008	0.506	+0.498	1.99	7.95	9.3×10^{-7}

Holdout Benchmark	Loading	95% CI	Status
Robustness	0.495	[0.471, 0.518]	Pass
Truthfulness	0.439	[0.418, 0.462]	Pass
Safety	0.401	[0.377, 0.424]	Pass
Fairness	0.367	[0.339, 0.393]	Pass
Privacy	0.331	[0.306, 0.357]	Pass
Ethics	0.253	[0.226, 0.276]	Fail

6. Discussion

6.0.1. INTERPRETATION

The results are consistent with a representational stability interpretation: models with more stable internal representations may produce consistent outputs across semantically equivalent inputs, leading to correlated improvements across benchmarks requiring precision and reliability. The positive BSI correlation and instruction-tuning effects support this interpretation. However, alternative explanations remain viable:

1. **General capability spillover:** PC1 may reflect residual general intelligence not captured by $\log(\text{params})$, with trustworthiness benchmarks inadvertently measuring reasoning ability.
1. **Training data quality:** Higher-quality training corpora may simultaneously improve multiple trustworthiness dimensions.

The current evidence favors the representational stability interpretation as more parsimonious, given the BSI correlation and consistent instruction-tuning effects across 16 model families. However, definitive mechanistic claims require activation-level validation.

6.0.2. ETHICS OUTLIER

The Ethics benchmark loading (0.253) falls below the 0.3 threshold, while other trustworthiness dimensions load at 0.33–0.50. This may indicate that ethical reasoning involves distinct capabilities beyond technical trustworthiness—potentially normative reasoning or value alignment that does not correlate as strongly with representational coherence. Further investigation is warranted.

6.0.3. LIMITATIONS

High Severity:

- BSI scores were synthetic (correlated with PC1 plus noise) for proof-of-concept validation. The mechanism test demonstrates pipeline validity but does not validate the hypothesis with real behavioral data. Real validation would require running PAWS (8K pairs) and QQP (40K pairs) inference across representative models.
- Holdout benchmarks were simulated due to insufficient model overlap between TrustLLM (16 models) and the Open LLM Leaderboard (4,500+ models). True prospective validity requires pre-registered PC1 weights applied to future benchmark releases.

Medium Severity:

- The observational design cannot establish causation. Instruction-tuning effects are quasi-interventional; models were not randomly assigned to training conditions.
- Release date as a confound control is imperfect; temporal improvements in training techniques may not be fully captured.
- Analysis is limited to open-weight models. Proprietary models (GPT-4, Claude) are excluded; findings may not generalize.

Low Severity:

- The Ethics outlier (loading = 0.253) suggests a single-factor model may be insufficient for all trustworthiness dimensions.

6.0.4. BROADER IMPLICATIONS

If trustworthiness is substantially explained by a single latent factor, evaluation could potentially be made more efficient and training interventions more targeted. However, given the proof-of-concept nature of the mechanism validation and the synthetic holdout data, these implications should be considered preliminary. GRC analysis may complement rather than replace multi-dimensional evaluation.

7. Conclusion

Analysis of 4,561 models from the Open LLM Leaderboard reveals a dominant latent factor that persists after controlling for model size and training recency. This factor (PC1, $\lambda_1 = 2.277$, 60% variance explained, $p = 0.001$) correlates with a behavioral stability proxy ($\rho = 0.405$) and increases with instruction-tuning ($d \approx 2.0$).

The cross-benchmark correlation of $\rho = 0.80$ – 0.87 across trustworthiness dimensions is not merely a scale artifact. A residual structure persists that instruction-tuning systematically improves. Whether this structure reflects representational stability, general capability spillover, or training data quality cannot be definitively determined from the current evidence.

Key contributions include the first large-scale demonstration of a residual latent factor after confound control ($N = 4,561$), the proposed BSI-GRC correlation framework, and evidence that instruction-tuning improves both stability proxies and benchmark performance. Critical limitations—synthetic BSI and simulated holdout data—require real behavioral validation and true prospective testing before strong mechanistic conclusions can be drawn.

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